# Problem Statement

## Prompt

We need a system to manage inventory. It should track products, stock levels, and allow updates when items are added or removed. As an Architect provide the efforts and major tasks needed for the development

Act as a Software Architect and analyze the requirements for an Inventory Management System.

- **Objective:** Manage products, track stock levels, and support stock additions and removals with validation and audit history.
- **Tasks:** Identify the major development tasks, architectural components, dependencies, and implementation sequence.
- **Effort estimation:** Estimate effort for each major task in person-days, stating key assumptions.
- **Project planning:** Recommend team roles, milestones, and an overall timeline.
- **Risks and gaps:** Highlight key technical risks, non-functional requirements, and missing information.

Present the results in structured tables, summarize the total estimated effort, and explicitly state assumptions. Keep the analysis at the architecture and planning level; avoid code implementation.

## Response

Here's a structured **high-level task breakdown** for an **Inventory Management System** that tracks products, stock levels, and stock updates.

# 1. Requirements & Planning

- Gather business requirements

- Define inventory workflow
- Identify user roles (admin, warehouse staff, manager)
- Define stock movement rules
- Finalize reporting requirements
- Decide deployment type (web/cloud/local)

# 2. Product Management Module

- Create product catalog
- Add/Edit/Delete products
- Manage product categories
- Add SKU/barcode support
- Store product details:
    - Name
    - Description
    - Price
    - Supplier
    - Unit type
    - Reorder threshold

# 3. Inventory / Stock Management

- Track current stock levels
- Add stock (purchase/restock)
- Remove stock (sales/damage/use)
- Transfer stock between locations
- Maintain stock adjustment logs
- Support batch/lot tracking (optional)
- Support expiry tracking (optional)

# 4. Stock Movement & Transactions

- Record all inventory transactions
- Maintain transaction history
- Track:
    - Inward stock
    - Outward stock
    - Returns
    - Damaged goods

- Add timestamps and user tracking

# 5. Supplier Management

- Create supplier records
- Manage supplier contact details
- Link suppliers to products
- Track purchase history

# 6. Purchase & Receiving

- Create purchase orders
- Receive inventory from suppliers
- Validate received quantities
- Update stock automatically

# 7. Sales / Dispatch Integration

- Deduct stock on sales
- Handle returns/refunds
- Generate dispatch records
- Prevent negative inventory

# 8. Alerts & Notifications

- Low stock alerts
- Out-of-stock alerts
- Expiry alerts
- Overstock alerts
- Email/SMS/in-app notifications

# 9. Search & Filtering

- Search products by:
    - SKU
    - Name

    - Category
    - Supplier
- Filter by stock status
- Sort inventory lists

# 10. Reporting & Analytics

- Current inventory report
- Stock movement report
- Low stock report
- Inventory valuation report
- Fast-moving/slow-moving products
- Export reports (PDF/Excel/CSV)

# 11. User & Role Management

- User authentication
- Role-based permissions
- Audit logs
- Password management

# 12. Database Design

- Design inventory schema
- Create product tables
- Create transaction tables
- Optimize indexing
- Setup backups

# 13. API / Backend Development

- Product APIs
- Stock update APIs
- Transaction APIs
- Authentication APIs
- Reporting APIs

# 14. Frontend / UI Development

- Dashboard
- Product management screens
- Stock update forms
- Inventory tables
- Reporting dashboards
- Mobile responsive UI

# 15. Security & Validation

- Input validation
- Prevent unauthorized access
- Secure APIs
- Data encryption
- Audit logging

# 16. Testing

- Unit testing
- Integration testing
- Inventory accuracy testing
- Performance testing
- User acceptance testing

# 17. Deployment & DevOps

- Setup hosting/server
- Configure database
- CI/CD pipeline
- Monitoring & logging
- Backup & recovery setup

# 18. Maintenance & Future Enhancements

- Barcode scanner integration
- QR code support

- Multi-warehouse support
- Mobile app
- AI demand forecasting
- ERP/accounting integration

If you want, I can also help with:

- converting this into a **project roadmap**
- creating **Agile epics and user stories**
- designing a **database schema**
- creating a **system architecture diagram**
- generating a **Jira task breakdown**
- building an **MVP scope** first
- suggesting a **tech stack** for web or mobile apps